

# ITO4132-TP6-25 Assessment 1

VFC Normalisation

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## UNF

CLASS (centre\_id, centre\_name, class\_num, class\_type, class\_desc, class\_fac\_room\_num, class\_startdatetime, class\_session\_count, (member\_num, class\_pay\_date, class\_session\_date, class\_attend\_count))

## 1NF

CLASS (centre\_id, centre\_name, class\_num, class\_type, class\_desc, class\_fac\_room\_num, class\_startdatetime, class\_session\_count)

**CK:**

- centre\_id, class\_num

CLASS\_ATTENDANCE (centre\_id, class\_num, member\_num, class\_pay\_date, class\_session\_date, class\_attend\_count)

**CK:**

- centre\_id, class\_num, member\_num, class\_session\_date

## Partial Dependencies:

centre\_id → centre\_name

centre\_id, class\_num, member\_num → class\_pay\_date

## 2NF

CLASS (centre\_id, class\_num, class\_type, class\_desc, class\_fac\_room\_num, class\_startdatetime, class\_session\_count)

CENTRE (centre\_id, centre\_name)

CLASS\_ATTENDANCE (centre\_id, class\_num, member\_num, class\_session\_date, class\_attend\_count)

CLASS\_ENROLMENT (centre\_id, class\_num, member\_num, class\_pay\_date)

## Transitive Dependencies:

class\_type → class\_desc

## 3NF

CLASS (centre\_id, class\_num, class\_type, class\_fac\_room\_num, class\_startdatetime, class\_session\_count)

CENTRE (centre\_id, centre\_name)

CLASS\_ATTENDANCE (centre\_id, class\_num, member\_num, class\_session\_date, class\_attend\_count)

CLASS\_ENROLMENT (centre\_id, class\_num, member\_num, class\_pay\_date)

CLASS\_TYPE (class\_type, class\_desc)

## Full Dependencies:

centre\_id, class\_num → class\_type, class\_fac\_room\_num, class\_startdatetime, class\_session\_count

centre\_id → centre\_name

centre\_id, class\_num, member\_num, class\_session\_date → class\_attend\_count

centre\_id, class\_num, member\_num → class\_pay\_date

class\_type → class\_desc

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### Attribute Synthesis

None.

### Final 3NF Relations

CLASS (centre\_id, class\_num, class\_type, class\_fac\_room\_num, class\_startdatetime, class\_session\_count)

CENTRE (centre\_id, centre\_name)

CLASS\_ATTENDANCE (centre\_id, class\_num, member\_num, class\_session\_date, class\_attend\_count)

CLASS\_ENROLMENT (centre\_id, class\_num, member\_num, class\_pay\_date)

CLASS\_TYPE (class\_type, class\_desc)